

Dissertation Structure

MSc Applied Social Data Science

This document outlines a structure for your dissertation based on your research proposal and literature review on the demographic transition fitness paradox. The structure follows a standard empirical social science dissertation format.

1. Title Page

- Dissertation title
- Your name
- Programme and institution
- Supervisor
- Submission date

2. Abstract (250–400 words)

- Research question
- Theory and motivation
- Data and methods
- Main findings
- Contribution

3. Acknowledgements (Optional)

- Short thank-you section

4. Table of Contents

- Table of contents
- List of tables
- List of figures
- Abbreviations

Chapter 1 — Introduction (~2,000–3,000 words)

- Background and motivation
- Research problem

- Research questions
- Hypotheses
- Contribution
- Dissertation structure overview

Chapter 2 — Literature Review (~4,000–6,000 words)

- Demographic transition literature
- Theoretical explanations
- J-curve literature
- Mechanism studies
- Research gaps
- Contribution of the present study

Chapter 3 — Data and Methodology (~3,000–5,000 words)

- Research design
- Datasets and sources
- Variable construction
- Methods for H1
- Methods for H2
- Methods for H3
- Methods for H4
- Robustness checks
- Ethical considerations

Chapter 4 — Descriptive Results (~2,000–3,000 words)

- Cohort fertility trends
- Cross-country patterns
- Education distributions
- Descriptive visualisations
- Preliminary evidence for inversion

Chapter 5 — Main Results: SES-Fertility Gradient and J-Curve (~3,000–5,000 words)

- Fixed-effects results
- Cross-country variation
- Quantile regression
- GAM / spline analysis
- J-curve evidence
- Robustness checks

Chapter 6 — Mechanisms Results (~3,000–5,000 words)

- Machine learning feature selection
- SHAP interpretation
- Bayesian hierarchical modelling
- Mediation analysis
- Cultural vs economic mechanisms

Chapter 7 — Discussion (~2,000–4,000 words)

- Summary of findings
- Theoretical implications
- Comparison with literature • Strengths and limitations
- Future research directions

Chapter 8 — Conclusion (~1,000–2,000 words)

- Restate research question
- Summarise findings
- State contribution
- Final theoretical implications

References / Bibliography

- Use consistent citation style
- Recommended tools: Zotero, Mendeley, BibTeX

Appendices

- Robustness tables
- Additional figures
- Variable definitions
- Model specifications
- Code and preprocessing pipeline
- SHAP plots and diagnostics
- Supplementary analyses